Handling Data 5 — Laws of probability · Independent events · Tree diagrams · Conditional probability

Marks in brackets. ★ = harder question. Independent events: $P(A \text{ and } B) = P(A) \times P(B)$. Mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$.

1 On a school day, the probability that Maya's bus is on time is 0.8 and the probability that her train is on time is 0.9. The two events are independent. Find the probability that [4]

a both are on time **b** exactly one of them is on time.

a $0.8 \times 0.9 = 0.72$. **b** Bus on time and train late, *or* bus late and train on time: $0.8 \times 0.1 + 0.2 \times 0.9 = 0.08 + 0.18 = 0.26$.

2 A year group of 120 students was asked whether they are left- or right-handed. [4]

	left-handed	right-handed	total
boys	7	58	65
girls	10	45	55
total	17	103	120

a A student is picked at random. Find the probability that the student is right-handed.

b Given that the student is a girl, find the probability that she is left-handed.

a $\dfrac{103}{120}$. **b** Only the 55 girls count now: $\dfrac{10}{55} = \dfrac{2}{11}$.

3 Each morning Leo tries to remember to take his water bottle to school, and each afternoon to bring it home. The probability that he remembers to take it on either journey is $\frac{3}{4}$. If he sets off with it, the probability that he loses it on that journey is $\frac{1}{6}$. On a particular day, find the probability that [10]

a he takes the bottle to school and loses it on the way

b he arrives at school with the bottle **c** he leaves school with the bottle

d he arrives home with the bottle **e** the bottle is at home in the evening.

Multiply along each chain of events.

a $\frac{3}{4} \times \frac{1}{6} = \frac{3}{24} = \frac{1}{8}$. **b** Takes it and does not lose it: $\frac{3}{4} \times \frac{5}{6} = \frac{15}{24} = \frac{5}{8}$. **c** It must be at school (b) and he must remember it: $\frac{5}{8} \times \frac{3}{4} = \frac{15}{32}$. **d** Leaves with it (c) and does not lose it: $\frac{15}{32} \times \frac{5}{6} = \frac{75}{192} = \frac{25}{64}$.

e Three ways: he never took it ($\frac{1}{4}$); he took it to school and forgot to bring it back ($\frac{5}{8} \times \frac{1}{4} = \frac{5}{32}$) — but then it is at school, not at home, so this does *not* count; or he brought it home (d). So $P = \frac{1}{4} + \frac{25}{64} = \frac{16}{64} + \frac{25}{64} = \frac{41}{64}$.

4 Ana, Ben, Cara and Dan run a race. Their probabilities of winning are shown. Separately, Eve and Finn play one game of chess; the probability that Eve wins is 0.6 (there are no draws). [7]

Ana	Ben	Cara	Dan
0.35	0.25	0.22	0.18

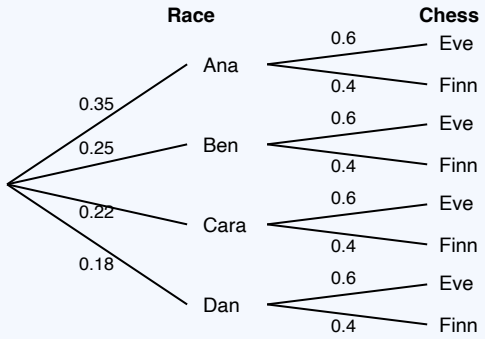
a Find the probability that Ana or Cara wins the race.

b Draw a tree diagram showing the winner of the race and the winner of the chess game.

c Find the probability that Ben wins the race and Finn wins the chess game.

a Only one person can win, so the events are mutually exclusive: $0.35 + 0.22 = 0.57$.

b Four branches for the race, each followed by two branches (Eve 0.6, Finn 0.4) for the chess:



c The race and the chess are independent: $0.25 \times 0.4 = 0.1$.

Total: 25 marks

Check: independent → multiply; mutually exclusive → add · "exactly one" = add the two one-and-not-the-other products · on a tree, multiply along a branch, add between branches · "given that" shrinks the total to the given row/column · check the probabilities on each set of branches add to 1.